HengSeng Sentiment Price Prediction agent Intern position assessment

The document gives a brief introduction to an assignment to check your qualifications for the internship as AI agents developer. Please complete it on your own time and submit it within 1 week.

Project Overview
Create a Python-based sentiment pipeline that predicts an asset's price at n+1 based on min. 2 sentiment sources. The system will scrape social posts, ingest recent news using exa.com, and train a predictive model using rigorous out-of-sample validation.

Technical Requirements
https://apify.com/ - for data scraping (alternative data features)
EXA.com for news sentiment

Optional social sources (you must use 2 different sources out of the list per each exchange):

- **LIHKG (Lihkg):** A dominant local community forum with dedicated stock and financial channels where Hong Kong retail traders actively debate market trends, local equities, and international tech stocks.
- **Telegram**
- **Futu / Moomoo Community:** The leading retail brokerage platform in Hong Kong, featuring highly active stock discussion boards integrated directly into trading feeds where retail investors discuss specific HK tickers in real time.
- **AAStocks & ETNet:** Major financial portals in HK that provide market data alongside user comment sections and discussion boards for individual stock quotes.
- **Naver Finance Stock Discussion Boards (종목토론실):** The absolute epicenter of retail investor sentiment in South Korea. Every single stock listed on the KRX has a dedicated Naver discussion room where retail investors share real-time sentiment, rumors, and ticker analysis.
- **Daum Finance Forums:** Another prominent financial portal providing active retail discussion spaces and community sentiment metrics for Korean equities.
- **KakaoTalk Open Chats & Telegram Channels:** Widely used across South Korea for closed-group, real-time peer discussions, trading alerts, and ticker-specific communities.
- **Blind:** A professional networking app frequently used in Korea where verified employees discuss company performance, giving unique insider sentiment on Korean enterprises and chaebols.
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Project Scope
Develop social sentiment price prediction model

1. Sentiment & Price Ingestion (Apify-based)
   - Scrape the social media posts about 5 different assets in each exchange (ticker, cashtag and keyword searches) for the last 10 days using Apify
   - Extract post timestamps (UTC alignment), post text, engagement (likes, reposts, views) and author follower counts, and align them with the price bars.
   - most recent and relevant news on the asset to complement the social sentiment (use exa.com)
2. extract per each ticker:
   - ticker, asset name, current price
   - sentiment, direction
   - position if any
   - support resistance levels

lets see your creativity here…

3. generate report sorting tickers by social mentions / sentiment score and direction per each exchange - visual report!
4. predict the price on n+1 and check your results

Submission by email gilad@crowdwisdomtrading.com including apify/tavily/exa

Submit the following:

1. A link to your GitHub/GitLab repository.
2. APIFY /exa tokens and actor IDs used for scraping (so we can reproduce your pipeline).

Timeline
You will have 5 - 7 days to complete this task. Please ensure you submit the deliverables by the end of this period.

We look forward to seeing your solution\!
Thanks for applying!
Gilad
CrowdWisdomTrading CEO